

Next-Generation AI-Enabled Incident Responses for Military Cloud Networks

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Abstract—We explore the feasibility of a next-generation AI-enabled automated incident response approach for military networks hosted on cloud infrastructure, utilizing infrastructure-as-code principles to enable rapid and scalable recovery mechanisms. The methodology includes isolating compromised components, preserving evidence for forensic analysis, and rebuilding affected resources to ensure the continuity of critical military services. The proposed approach provides a resilient, automated, and scalable solution to maintain critical service availability during cyber incidents. The experimental results are analyzed to validate the effectiveness of the methodology in real-world military network scenarios. We describe the challenges faced while executing incident response in the cloud. We also discuss the integration of autonomous cyber defense agents for automating incident detection and response. The paper outlines the possible further implementations of decoys, transforming the environment into a continuous training ground for the red team and eventual AI vs. AI playground.

Index Terms—Cloud Incident Response, Military Networks, Tactical Edge

I. INTRODUCTION

NATO and coalition members are increasingly adopting cloud systems to improve scalability, operational flexibility, and collaboration in geographically dispersed environments. At the same time, due to geopolitical dynamics, military networks are exposed to a growing number of threats. These conditions raise the critical need for innovative and robust incident response mechanisms tailored to the unique operational and security requirements of NATO and allies' military networks.

The cloud incident responses differ from those executed in on-premise infrastructures. Cloud environments enhance incident responses by enabling the scalable deployment of response resources, facilitating easy automation, and providing built-in advanced security tools. Infrastructure-as-Code (IaC) principles implemented in cloud services open the possibility of rapidly decreasing the reaction time, while increasing the flexibility and automation of each response. This paper focuses on incident response in military-owned cloud infrastructure. In contrast to the shared responsibility model used in public cloud computing, in a military cloud infrastructure, all responsibilities related to incident responses must be accounted for by military personnel managing the infrastructure and cannot be shared with Cloud Service Provider (CSP). Although CSP provides essential security

services and tools, the response to military cloud incidents must remain under the direct control of military defense operators to detect, respond to, and mitigate threats within their cloud environment. While CSP offers foundational security measures, they are not directly involved in mission-specific operations. CSP are not involved in the situational awareness of the military operation and cannot participate in the decision-making process in such contexts. In our research scenario, despite all the actions taken to prevent a cybersecurity incident, such as proper logging, monitoring, and boundary protection, we assume that there is evidence of a breach, and the containment phase of the incident response plan is activated. Previous research on this topic highlights the evolving area of cloud incident responses [1], [2], where traditional methods are being adapted and supplemented with cloud-native approaches to address specific challenges [3]. Several incident response frameworks have been proposed to unify cloud incident response practices and provide tailored guidance to cloud users and service providers, addressing both technical and operational aspects of incident response [4], [5]. Our research fits into and combines the third phase of the incident response model of National Institute of Standards and Technology (NIST) Special Publication (SP) 800-61 (Revision 2), 'Containment, Eradication, and Recovery' and the elements 'Preparation' and 'Automation and orchestration' of the Cloud Security Alliance (CSA) approach [6] to provide a specialized framework for managing incidents in cloud environments.

II. NEXT GENERATION INCIDENT RESPONSE

The primary objectives of the multilevel automated incident response are incident containment, rapid recovery, and maintenance of mission-critical service availability. The design of the responses is based on a mapping between them and a matrix of real-world tactics and techniques employed by cyber adversaries across various mission network topologies. The proposed pipeline strikes a balance between automation and oversight. Routine incidents are handled autonomously, while critical cases benefit from human validation.

The various types of incident responses align with the tactics and techniques outlined in MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) Matrix [7] for enterprises relevant to cloud systems.

Instance-level responses mitigate tactics such as Impact Defense Evasion, Impact, and Credential Access, quickly remediate compromised resources, and effectively wipe the attacker’s foothold. This level of response addresses techniques such as Data Destruction and Resource Hijacking by eliminating and recreating compromised instances. The response limits the risks of malware persistence and potentially mitigates risks related to the Credential Dumping technique by isolating instances before attackers can pivot.

Subnet-level response focuses on Lateral Movement and Persistence tactics within the subnet boundaries. Taking a snapshot and redeploying all instances in a compromised subnet limits adversary lateral movement within the network. The subnet-level response extends the scope of remediation by addressing lateral-movement techniques, such as Remote Service Exploitation or Internal Spearfishing.

AZ-level response adds coverage of Impact, Command and Control (e.g., rerouting network traffic), Privilege Escalation, and Defense Evasion tactics. It can help when subnets or Availability Zone (AZ)s are targeted in distributed attacks. Shifting workloads to an alternate AZ disrupts adversaries’ attempts to maintain control over compromised infrastructure.

A. Decision Making

Traditionally, incident response has relied on static workflows or human operators in Security Operations Center (SOC), but the increasing scale and complexity of adversarial activities require a more adaptive, AI-driven approach that integrates cloud-native monitoring, large language models (LLM), automation through serverless computing, and human oversight through SOC escalation.

To implement a next-generation cloud incident response, we propose a novel Universal Decision Making Framework (UDMF) as a methodological contribution to incident response. UDMF integrates automation and human oversight to ensure efficient, scalable, and context-aware decision-making in diverse operational environments. This decision-making can be further enhanced by intelligence enrichment methods that combine named entity recognition, TTP extraction, and knowledge graph reasoning to provide the agent with a more contextually accurate representation [8]. It is designed to be universally applicable and to adapt to various cloud infrastructures and security postures. The universal nature of this framework enables its application across Public, Private, and Hybrid cloud environments. However, its implementation may require varying levels of resource investment depending on the infrastructure.

The proposed AI-driven decision flow operates as a pipeline composed of five stages. The **triggering layer** continuously monitors infrastructure telemetry for abnormal patterns such as repeated authentication failures, unexpected network flows, or deviations in system logs. Once the configured thresholds are exceeded, the layer generates an event that initiates the decision process. The **evaluation layer** leverages generative AI models to parse the description of the incident, correlate it with the stored threat intelligence

and historical attack data, and classify the severity of the event. Unlike traditional rule-based systems, this AI-driven layer can adaptively interpret novel indicators of compromise (IoCs) and recommend appropriate mitigations. The decision flow must be context-sensitive. This requires combining infrastructure-level representations (e.g., terraform scripts capturing topology and dependencies) with operational awareness from situational awareness tools (e.g., NCOP, ATAK), which map network segments to specific missions or units. The Universal Decision Making Framework is designed to ingest these two dimensions, ensuring that automated or AI-generated playbooks do not disrupt mission-critical operations. The **execution layer** selects or synthesizes a mitigation workflow based on the LLM evaluation and executes the selected response directly in the cloud environment. This may involve invoking predefined playbooks (e.g., isolating an instance, revoking compromised IAM credentials) or dynamically generating new scripts using Terraform, Ansible, or Python. The **notification layer** triggers a notification for each action executed. These notifications ensure that the SOC maintains full situational awareness of every triggered response, whether automated or AI-generated. All events are logged and version-controlled for auditability. The **escalation layer** activates the SOC involvement at one of two levels, depending on the incident severity assigned by an LLM. For low and medium-severity events, SOC analysts are just notified about the response to maintain awareness, but no approval is required (*inform-only escalation*). For high-severity or ambiguous events, the execution layer pauses execution of the response and requests SOC approval before deploying it (*approval-based escalation*). This ensures that human oversight governs mission-critical decisions, preventing erroneous automation in sensitive contexts. It should be noted that in operational environments, the AI decision flow could be designed to comply with strict security policies (e.g., DoD, NATO Cyber Policy), and thus may restrict or delay certain automated responses.

In an infrastructure integrated with Autonomous Cyber Defense (ACyD) agents, decisions to execute incident responses are driven by the agent’s ability to detect abnormal behavior using pre-trained models and threat intelligence databases [9]. This approach relies on the adequate agent’s training. The agents operate independently by continuously monitoring the infrastructure for Indicators of Compromise (IOC) and triggering pre-scripted responses [10]. Using ACyD enables a real-time response without human intervention, making it ideal for large-scale cloud infrastructures or environments with a low tolerance for delay. As a drawback of this approach, we identified the need to regularly retrain the agents in non-production environments to keep them adaptive to new or complex threats. This continuous training could lead to significant costs and introduce some security challenges due to possible adversarial AI attacks. Additionally, the pre-scripted playbooks must be regularly updated to reflect changes in the cloud infrastructure. The introduction of a

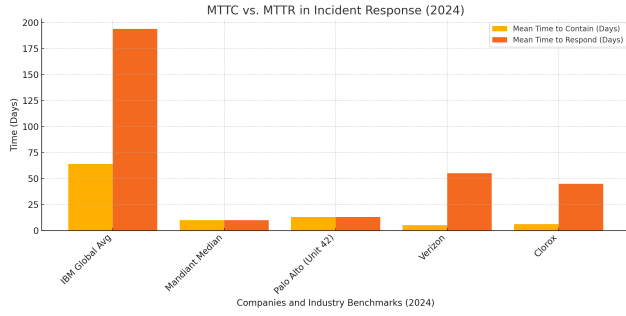


Fig. 1: Industry MTTR and MTTC values in 2024, according to cybersecurity reports.

blue agent capable of dynamically generating new incident response playbooks marks a significant evolution from the more static automation approaches. Without the blue agent, responses are largely predefined by automation scripts, which means that each possible attack vector must be anticipated in advance. This approach, while effective for common threats, proved less adaptive when faced with novel indicators of compromise. A blue agent that uses generative AI to synthesize new mitigation strategies in real time can dynamically evaluate whether a new response is warranted. For example, if lateral movement is detected in a subnet that had not been part of prior playbooks, the agent can author a Terraform snippet that isolates the compromised subnet or generate an IAM policy that restricts permissions until remediation is complete. The blue agent continuously observes the telemetry and correlates the logs with the current state of the deployed infrastructure. When a new IOC is identified, it converts this knowledge into executable code. The use of generative AI services, such as the models in AWS Bedrock, enables the agent to produce syntactically correct and context-aware configurations across multiple scripting languages (Terraform, Python, Ansible). Each generated response should be stored in a version-controlled repository, where it should undergo policy compliance checks before activation. This will ensure that automation does not bypass organizational governance or the NIST CSF functions of govern, identify, and protect.

Dynamic scripting capacity provides defenders with resilience against zero-day exploits and rapidly evolving attack vectors. Instead of waiting for human analysts to design and test new mitigations, the blue agent shortens the breach lifecycle by producing executable defenses in minutes, based on a limited set of actions. Such adaptability is particularly relevant in military cloud environments, where maintaining mission continuity is crucial despite adversarial attempts to degrade infrastructure. By combining automated detection with AI-driven generation of new defensive scripts, the system shifts from reactive containment to proactive adaptation. This represents not just an operational enhancement, but a conceptual step toward autonomous cyber defense ecosystems capable of self-healing and continuous learning.

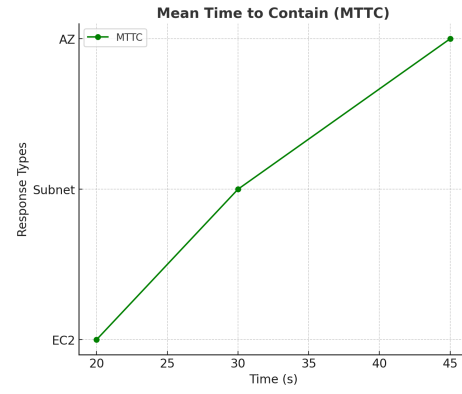


Fig. 2: The Mean Time to Contain (MTTC) metric: time passed from triggering the response to the destruction of the affected instances.

B. Response efficiency

Incident responses should be accompanied by performance metrics and sufficient information to analyze results and identify potential future improvements. Effective and immediate incident responses are essential for military networks. Ensuring operational continuity, protecting sensitive data, and ensuring operability through different platforms are critical for decision-makers in these environments. In the following, we discuss several possible metrics that can be used to measure the efficiency of responses and find the balance between cost and efficiency.

The **response times** are critical in operational networks, where fast containment of incidents ensures operational continuity and limits adversarial impacts. The proposed multi-layered hierarchical architecture enables efficient response execution across all levels, i.e., Instance, Subnet, and AZ, by utilizing pre-scripted and automated workflow. We used two time-related metrics to measure the effectiveness of the responses [11]: Mean Time to Contain (MTTC) and Mean Time To Respond (MTTR) [12]. These metrics are widely considered critical incident response key performance indicators (KPIs) [5], [13]. Reducing MTTC and MTTR requires a combination of specialized tools and processes to improve incident containment and remediation efficiency. Moreover, MTTC is a key metric, as it reflects the overall effectiveness of an incident response team. At the same time, MTTR measures the time required to restore systems to an operational state. In our experimental environment, this encompassed the time between triggering the response script and the successful rebuilding of the infrastructure, including the restoration of normal operations for the affected instances and applications. MTTR represents the time elapsed from the incident's detection to its resolution. MTTC, on the other hand, corresponds to the time between incident detection and incident containment. In the context of the described responses, this will include only the time from detecting the incident to destroying the compromised machines. The MTTC and MTTR values appear in increments because the controlled experimental environment used fixed orchestration

steps. Minor fluctuations were rounded to the nearest 5–10 seconds for clarity.

The **cost of response** is optimized by employing IaC and automated orchestration tools. Utilizing dynamic on-demand provisioning of resources in the cloud and cost-effective, cloud-native tools to execute responses, the system balances cost efficiency with the robustness required in military information systems infrastructures. The cost of each of the responses reflects the cost of the infrastructure that is affected, e.g., a single instance, multiple instances, or an AZ. Taking snapshots during incident responses ensures not only forensic capabilities, but also prevents doubling the cost, as would be the case if we isolate the compromised instances and keep them alive. For the experimental environment, as described in Section III-A, the cost is less than 700\$ per month, with the condition that this is a static experimental environment without inbound or outbound traffic generation. The **availability** of the mission-critical military applications is measured during the execution of incident responses. This metric is used to fine-tune the execution times and the architecture of the reactions. The architecture is designed to restore the availability and accessibility of the applications after an incident response process. By isolating and recreating affected resources in parallel, while rerouting traffic to unaffected components, operational continuity is ensured. Specific strategies are implemented at the Subnet-level responses to safeguard the infrastructure hosting the critical military application from Denial of Service (DOS) and Distributed Denial of Service (DDOS) attacks. This is achieved during the re-creation actions by dynamically changing the physical location of the Elastic Compute Cloud (EC2)s hosting the application to another cloud region and dynamically re-routing the inbound and outbound traffic related to that service, guaranteeing continuity of the critical military application.

It is important to note that in our experimental environment, and given the sensitivity of operational data, we assume that mission data is not stored locally on cloud instances. Instead, data is retrieved through APIs or other secure mechanisms from trusted repositories such as data warehouses or data lakes, thereby reducing the risk of data staleness or inadvertent restoration of compromised content. The proposed responses need to be **CSP-agnostic**, enabling easy integration with different providers or cloud management orchestration tools. By relying on the IaC principles and open Continuous Integration & Continuous Delivery (CI/CD) frameworks, the responses can be deployed in heterogeneous environments, ensuring flexibility and compatibility. This type of adaptability provides readiness across various military cloud and edge infrastructures.

III. VALIDATION ENVIRONMENT

To validate our concept of next-generation incident response, we have utilized a virtual laboratory provided by the NATO Allied Command Transformation (ACT) Science and Technology Experimentation Environment

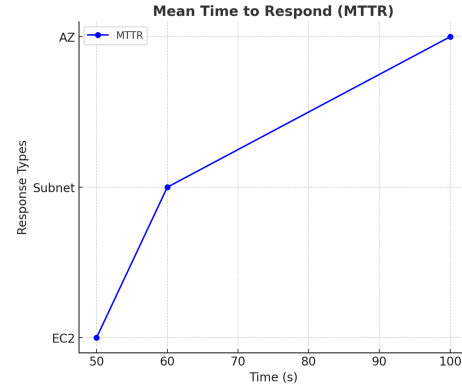


Fig. 3: The Mean Time to Respond (MTTR) metric: time passed from triggering the response until its successful completion.

(SnTEE), which facilitates the transfer of scientific and technical activities conducted by the NATO Science and Technology Organization (STO) research task groups to NATO operations [14]. To build, change, and version the infrastructure on Amazon Web Services (AWS), we used Terraform scripts executed through the CI/CD pipeline provided by the virtual laboratory. Incident responses are assembled from scripts written in Terraform, Python, and AWS Command Line Interface (CLI) combined by Ansible playbooks. Additionally, to demonstrate a resilient, monitored, and automated cloud environment suitable for mission-critical applications, we have implemented native AWS services to ensure that the automation of the incident response workflow is done with high continuity and availability: Amazon CloudWatch, AWS Lambda, Amazon Simple Notification Service (SNS), Amazon Relational Database Service (RDS), AWS Amplify, AWS Web Application Firewall (WAF), and Amazon Bedrock.

A. Network Topology

The chosen network topology was inspired by Cyber Operations Research Gym (CybORG) Cyber Autonomy Gym for Experimentation (CAGE) challenge 4 [15], adapted to a cloud-based infrastructure. It consists of three AZs within a Virtual Private Cloud (VPC). In two of the AZs, the subnets host only mission operational EC2 instances. The first AZ hosts a dockerized JCHAT in an EC2 instance in the public subnet. Using Openfire as the underlying XMPP server for secure operational communications. The second AZ implements a web application hosting environment that includes three EC2 web servers, which are publicly accessible. The third AZ hosts an unclassified yet mission-critical military application. This AZ contains a private subnet that hosts the application's database instance, implemented using Amazon RDS (Fig. 4), which serves as the primary data repository for the application. Every EC2 in the architecture is integrated with CloudWatch monitoring capabilities that detect and respond to security threats through automated incident response mechanisms.

An automated incident response system features three levels of response capabilities, i.e., instance, subnet, and

availability zone levels. The system automatically terminates compromised instances, creates forensic snapshots, and launches replacement instances with identical configurations while preserving user data. The incident response system is triggered by CloudWatch alarms that monitor failures and attacks. An example of an attack is an SSH brute force.

The mission-critical communication and situation awareness services implemented in the infrastructure include a real operational tool called Joint Tactical Chat (JCHAT). Developed and deployed as a secure web-based chat capability, JCHAT provides real-time text communication and information sharing among distributed operators. The service is hosted as a Docker image on an EC2s instance within the public subnet of the first AZ, ensuring redundancy and availability. The availability of these services during the execution of the incident response is measured as one of the Key Performance Indicators (KPI) to evaluate the effectiveness of the response.

The incident responses investigated in our work are intended for cloud environments. Although the characteristics of next-generation cloud incident responses can be applied to hybrid and on-premise infrastructures, this comes with implementation constraints. The hybrid cloud includes a

relies on predefined images and baselines that can be adapted by infrastructure administrators, allowing engineers to update Amazon EC2 baselines and prevent reintroduction of vulnerabilities after recovery. In our current implementation, an instance-level response is initiated when failed SSH login attempts exceed two per minute on a single EC2 instance. A subnet-level response is activated when multiple instances within the same subnet exhibit correlated suspicious behavior, and an availability-zone-level response is triggered when security incidents propagate across several subnets within a single Availability Zone. Each Lambda function executes Python scripts that perform forensic snapshots, terminate compromised instances, and provision replacement instances with identical configurations, ensuring preservation of user data and Identity and Access Management (IAM) profiles.

Both metrics and the time results we achieved are represented in Fig. 3 and Fig. 2. The values are compared with the results reported by industry cyber security reports in 2024 [16]–[21], depicted in Fig. 1, that represent conventional approaches without AI. Our results show that the AI-enabled pipeline achieves notably lower MTTR and MTTC.

IV. OPEN CHALLENGES AND FUTURE DIRECTIONS

Introducing next-generation incident responses poses challenges in transforming traditional approaches to cloud administration and risk management, which must be considered.

The use of predefined incident responses alters the traditional way of cyber defense, which depends on the risk a defender is willing to take. When the number of compromised instances within a subnet exceeds a predefined threshold, an orchestration tool or autonomous agent escalates to a higher-level response, executing actions at a broader scale. Integrating predefined scripted responses enables interaction between cybersecurity engineers and the scripts or baseline configurations used to recreate an instance. This approach aims to automate response processes while preserving the critical role of the human factor in cybersecurity operations. Security engineers can tailor their responses to better align with the military’s decision-making processes. Human defenders will remain focused on monitoring the execution of automated responses, verifying their efficacy, and conducting further cybersecurity investigations.

A significant challenge in implementing automated responses is the risk of inadvertently reintroducing an incident when recreating a compromised instance. An adversary who gains knowledge of incident response mechanisms might deliberately trigger responses in ways that could cause disruptions. Incident response decisions based on predefined frameworks (such as MITRE ATT&CK) could incorrectly classify events, leading to unnecessary resource destruction or the failure to contain an actual attack. Certain functionalities provided by CSP might introduce vendor dependencies that limit portability across different military cloud environments. Large-scale responses (e.g., at the Software-Defined Networking (SDN) level) could

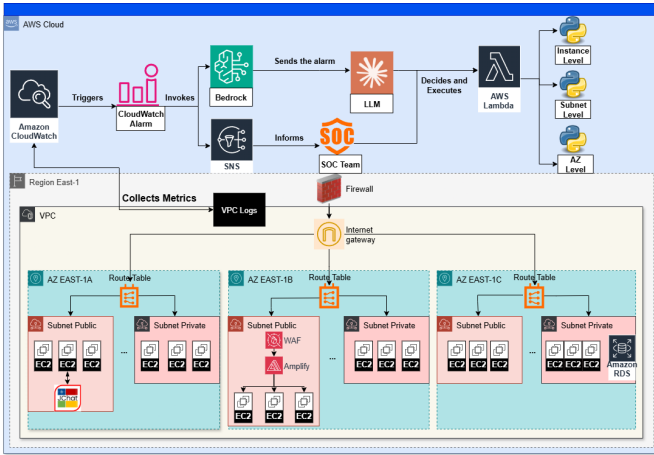


Fig. 4: Network topology describing the cloud experimental environment built on Amazon Web Services

combination of on-premise infrastructure and cloud resources, which requires responses to span their actions across all environments. Such a situation will involve orchestration and monitoring tools that can operate simultaneously on-premise and in the cloud. The response scripts must include additional scripting to handle both cloud and on-premise virtual snapshots simultaneously. The responses will have different levels of success in both environments, which will complicate decision-making and risk management. Although the core principles of next-generation cloud incident responses can be applied across different topologies, their implementation in non-cloud environments faces constraints, and their efficiency decreases, as shown in Table I.

Terraform scripts implementing the responses are maintained in a version-controlled Git repository. Each response

TABLE I: Comparison of Response Efficiency in Different Topologies

	Cloud	Hybrid	On-Premise
Orchestration Tools	Cloud-native automation	Multi-platform (e.g., Terraform, Kubernetes)	Local tools (e.g., Ansible, Puppet)
Scaling	Dynamic scaling of resources	Limited to available resources in both environments	Fixed capacity based on hardware limits
Snapshots	Cloud snapshots with minimal effort	Mix of cloud and local virtual snapshots	Only local virtual snapshots
Monitoring	Cloud-native monitoring tools	Unified monitoring with cloud and on-prem	On-premise monitoring platforms
Response Speed	High due to automated cloud capabilities	Variable depending on integration	Slower due to manual or limited automation

take several minutes, potentially leaving mission-critical applications vulnerable before containment is fully achieved. Rapid destruction and re-creation of instances can introduce temporary service outages, which could affect real-time military operations if not carefully planned.

To mitigate the listed drawbacks and cybersecurity concerns, a thorough implementation of layered authentication and access controls is necessary to prevent unauthorized modifications to automated responses. Introducing the human factor within the mechanisms for high-risk responses can ensure military oversight in critical scenarios. In addition, regular monitoring of the incident response KPIs should be performed to optimize response execution times and ensure minimal disruption of mission-critical applications.

A. Risk management

Risk management is one of the few aspects that differ significantly between military and industry-owned cloud infrastructures. Although both sectors aim to effectively manage risk in the cloud [22], their approaches differ in terms of focus, complexity, and specific standards.

The industrial decision-making process is more decentralized, focusing on mitigating financial losses, and risk tolerance can vary according to business objectives. In the event of cybersecurity incidents, companies often need to involve third-party entities, such as insurance companies, external cybersecurity experts, legal advisors, and CSPs, who can influence the incident response. Industry-owned cloud infrastructures adhere to cloud risk management standards and frameworks, such as [23]–[27].

In a military cloud infrastructure, risk management emphasizes national security, operational continuity, and adherence to strict protocols. While industry infrastructures emphasize minimizing downtime cost-effectively to ensure customer satisfaction and financial stability, military infrastructures focus on continuity under extreme conditions, prioritizing mission-critical tasks over cost. Military systems also prioritize the resilience and protection of classified data. It is possible to observe a complete infrastructure shutdown for the sake of damage control, an action that not many industries can afford to take. Military networks have to comply with the Alliance standards and frameworks (and in some cases also with national standards), such as [28]–[33].

B. Deception techniques

In the event that a threat actor advances quickly through compromised instances and subnets, a possible novel incident response approach could be to dynamically create additional instances to increase the number of potential targets for

the attacker, thus giving cyber-defenders time to contain the breach or protect sensitive data. Such an approach could be a last resort in high-stakes military environments where operational continuity and the preservation of sensitive data are critical and an integral component of a layered defense strategy.

Automated honeypot implementations could add flexibility to the incident response process. If Indicators of Compromise (IOC) are received from multiple instances within a subnet, the affected instances could be recreated in another subnet to provide operational continuity without destroying the compromised instances. The compromised instances could be used as a honeypot environment without disrupting the threat actor’s activities. This could provide valuable evidence for the threat actor Tactics, Techniques, and Procedures (TTPs). Additional measures should be taken to isolate the subnet from the rest of the environment and ensure that sensitive data cannot be exfiltrated from affected machines. Using IaC infrastructure and based on operational requirements, the military cloud infrastructure could also potentially use honeypot environments that intentionally attract sophisticated threat actors. An indication of compromise in such an environment could trigger an automated response that generates additional predefined infrastructure to keep the attacker engaged while analysis of its TTPs is in progress. This ensures that while the attacker keeps control of the original subnet, their actions are limited to interacting with the honeypot systems only.

V. CONCLUSIONS

In this paper, we have shown how IaC principles and automated orchestration tools can improve resilience, scalability, speed, and efficiency of incident response. The hierarchical approach to responses ensures rapid containment and recovery mechanisms while maintaining the availability of mission-critical applications. The experimental results reveal significantly better performance in terms of MTTR and MTTC metrics compared to publicly reported benchmarks for industry cloud networks. During our investigation, we analyzed key challenges related to the implementation of response, including the possibility of reintroducing incidents, adapting risk management to military-specific operations, and balancing automation with human oversight. While decoys and honeypots are well-known concepts, the automation of their deployment as part of an incident response strategy represents an innovative approach. Integrating automated decoy-based responses and dynamic honeypot formation offers possibilities for implementing new techniques to incident mitigation and intelligence gathering. Although our

next-generation AI-enabled incident response offers numerous advantages, enhanced protection should be ensured against vulnerabilities in automation tools, insider threats, and adversarial access to the response mechanism.

The next step in our research is the development of a red agent capable of executing adversarial tactics such as privilege escalation, lateral movement, and persistence, providing a realistic and continuously evolving challenge for the blue agent. In parallel, the blue agent would be tasked with dynamically adapting its defensive strategies in real time, expanding beyond predefined playbooks to counter the threats. Future research will focus on deploying this adversarial interaction within a real cloud environment, thereby creating a controlled playground where both offensive and defensive AI agents can operate autonomously. An important outcome of this work will be the generation of detailed datasets capturing these agent interactions, which can serve as a valuable resource for advancing autonomous cyber defense research and for training future models.

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